Supplementary Material

Two-Time-Scale Coordination of Vehicle-to-Grid Voltage Support and Frequency Reserve in a Two-Area System Using Proximal Policy Optimization

S1. Reproducibility

All results are generated by the released code from the raw simulation records. The repository (`https://github.com/kenoto99/v2g-ppo-two-area`, release v3.0.1; Zenodo DOI 10.5281/zenodo.23851585) contains `REPRODUCE.md` with the ordered commands, `results/ENVIRONMENT.txt` with the exact package versions, `results/CHECKSUMS.sha256` with SHA-256 checksums of every raw result, table, figure, trained policy and PDF, and `results/verification_report.md`, the output of `scripts/verify_all.py`. That script re-derives every number in the manuscript independently of the script that builds the tables and compares the two; it also re-runs the pandapower verification and the physics tests and checks that every trained policy carries the final model version tag (`env_version = v3`).

Random seeds. PPO training seeds 0 to 4 (main), 0 to 2 (ablation, squashed policy), 0 and 1 (weight sensitivity). Training-day generator `default_rng(1000 seed + env)` for four parallel environments; validation days `default_rng(10000)`; test days `default_rng(2026)`; stress days `default_rng(777)`; the disturbance of each hour `default_rng([scenario seed, hour])`, identical for every controller and treatment.

Package versions. Python 3.13, NumPy 2.4.6, SciPy 1.18.1, PyTorch 2.15.0 (CUDA build, executed on CPU), Gymnasium 1.4.0, Stable-Baselines3 2.9.2, pandapower 3.5.6, numba 0.68.1, matplotlib 3.11.2.

S2. Learning curves of all runs

Table 1 lists every training run with its final validation cost, the change of the validation cost over the last 50 000 steps (convergence criterion 2%), and wall-clock time on one CPU core.

Table 1: All training runs (final model version v3).

Run	Steps	Reward config	Final val. cost	Best val. cost	Change, last 50k (%)	Wall (min)
ppo_main_s0	400k	nominal	136.8	136.6	0.1	27
ppo_main_s1	400k	nominal	138.3	138.3	1.0	27
ppo_main_s2	400k	nominal	136.8	136.8	0.8	36
ppo_main_s3	400k	nominal	136.5	136.4	0.0	36
ppo_main_s4	400k	nominal	140.2	139.6	0.2	27
ppo_noDeg_s0	200k	'w_deg': 0.0	137.5	135.8	0.4	12
ppo_noDeg_s1	200k	'w_deg': 0.0	155.4	155.4	5.3	12
ppo_noDeg_s2	200k	'w_deg': 0.0	138.2	137.1	3.1	12
ppo_noF_s0	200k	'w_f': 0.0, 'w_tie': 0.0	81.0	80.2	0.4	13
ppo_noF_s1	200k	'w_f': 0.0, 'w_tie': 0.0	81.3	81.3	0.7	14
ppo_noF_s2	200k	'w_f': 0.0, 'w_tie': 0.0	81.8	80.0	0.8	12
ppo_noOV_s0	200k	'w_ov': 0.0	143.4	142.8	7.8	29
ppo_noOV_s1	200k	'w_ov': 0.0	153.8	153.8	9.0	22
ppo_noOV_s2	200k	'w_ov': 0.0	144.3	143.8	11.4	14
ppo_ref200_s0	200k	nominal	143.3	142.1	4.9	25
ppo_ref200_s1	200k	nominal	161.9	161.9	4.0	29
ppo_ref200_s2	200k	nominal	144.7	144.2	10.8	23
ppo_squash_s0	200k	nominal, squashed	178.2	177.7	0.2	15
ppo_squash_s1	200k	nominal, squashed	178.5	178.2	0.4	14
ppo_squash_s2	200k	nominal, squashed	177.9	177.5	0.0	27
ppo_wF2_s0	200k	'w_f': 2.0	119.3	119.3	4.8	19
ppo_wF2_s1	200k	'w_f': 2.0	126.2	126.2	0.7	12
ppo_wF8_s0	200k	'w_f': 8.0	219.6	218.3	0.4	12
ppo_wF8_s1	200k	'w_f': 8.0	217.7	217.7	4.3	12
ppo_wV100_s0	200k	'w_v': 100.0	213.5	213.0	0.0	12
ppo_wV100_s1	200k	'w_v': 100.0	213.2	213.2	1.7	19
ppo_wV25_s0	200k	'w_v': 25.0	114.4	114.2	0.1	12
ppo_wV25_s1	200k	'w_v': 25.0	113.2	113.0	1.8	12

S3. Model verification details

Base case of the IEEE 33-bus feeder: minimum voltage 0.9131 p.u. at bus 18, loss 202.68 kW. Largest voltage difference from pandapower over 20 random operating points: 1.2×10^{-10} p.u. Two-area LFC without AGC after a 0.02-p.u. step: $\Delta f = -0.02381$ Hz (theory -0.02381), $\Delta P_{\text{tie}} = -5.000$ MW (theory -5.000). The physics tests in `tests/test_physics.py` additionally verify the regulation-energy integral against a closed-form case, the feasibility of random (including out-of-range) actions on 40 train and stress days with the 60-s and the exact 3600-s fast layer, that no upward reserve is credited when driving alone takes the fleet below its floor, and that raw-action clipping and physical projection are counted separately.

S4. Regulation-energy treatments, per controller

Table 3 of the main text gives the daily results under each treatment. The tracking statistics against the exact 3600-s simulation on the test days are: gross energy ratio (treatment / exact) 2.9 (hourly extrapolation), 0.94 (decomposed) for PPO; 2.7 and 0.94 for RB; 2.6 and 0.94 for DAO. Net daily energy (MWh): PPO -0.73 exact, -0.69 decomposed, -1.67 extrapolated, -0.03 window; RB -0.50, -0.46, -1.10, -0.02; DAO -0.47, -0.43, -1.03, -0.02. Share of hours in which the extrapolation has the wrong sign: PPO 39%, RB 43%, DAO 43%. Fraction of simulated time at the droop saturation limit in the exact simulations: PPO 0.00%, RB 0.00%, DAO 0.00%. Minimum sustained-activation margin over test and stress days: PPO 0.000 MWh, RB 0.701 MWh, DAO 0.000 MWh.

S5. Raw-action saturation of the final policies

Share of hourly decisions above $+1$ / below -1 before clipping (test days, five seeds): a_1 13.4% / 25.1%, a_2 21.6% / 33.9%, a_3 43.2% / 10.8%, a_4 30.7% / 17.9%. Median excess beyond the bound when saturated: 0.61.

S6. Remaining model assumptions

Each area is represented by 25 identical copies of one IEEE 33-bus feeder; the fleet of each feeder is one aggregated battery with a single SoC; charger reactive power is not used; the LFC model is linear with representative parameters; load, PV and availability profiles are synthetic and shaped after published measurements; hour-ahead forecasts are assumed exact; the fast layer of each hour is a 60-s window containing one step event and a fluctuating component, and the regulation energy of the remaining 59 minutes is obtained from a second fluctuation-only window (checked against exact 3600-s simulations); the throughput term is a linear proxy for degradation; the weights of the reward were chosen by the authors and their effect is shown in the ablation and sensitivity studies.